

Object Oriented Programming In Java

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Introduction

- ``static`` belongs to the class, not the object
- Accessible without creating an object
- Memory allocated once per class

Static Variables

- Also called class variables
- Shared among all objects of the class
- Memory allocated only once
- Example: `static String college = "ABC College";`

Static Methods

- Belong to the class, not object
- Can be called without creating an object
- Cannot access non-static (instance) members directly
- Example: `MathUtil.square(5);`

Static Block

- Used for initialization of static variables
- Runs once when the class is loaded
- Example: `static { count = 100; }`

Static Nested Class

- Only nested classes can be declared static
- Access without creating outer class object
- Example: `Outer.Inner obj = new Outer.Inner();`

Real-Life Example

- `main()` method in Java is static
- Reason: JVM can call main without creating an object
- Example: `public static void main(String[] args) { ... }`

Static & Inheritance – Variables

- Static variables are inherited
- Shared at class level
- Changing in child reflects in parent too

Static Methods in Inheritance

- Static methods are inherited but cannot be overridden
- If redefined in child → method hiding
- `Parent.show(); Child.show();`

Why Static Methods Cannot Be Overridden

- Static belongs to class, not object
- Overriding works with runtime polymorphism
- Static resolved at compile-time → method hiding

Static Import – Introduction

- Allows accessing static members directly
- No need to prefix with class name
- Example: `import static java.lang.Math.*;`

Normal Import vs Static Import

- Normal import → imports classes
- Static import → imports static members
- Example: `import static java.lang.Math.PI;`

Advantages & Disadvantages

- Advantages: Cleaner code, useful for constants
- Disadvantages: Reduces readability, name conflicts possible

Example with Constants

- `import static java.lang.Math.*;`
- `System.out.println(PI);`
- `System.out.println(pow(2,3));`

Summary

- ``static`` → class-level keyword
- Saves memory by sharing variables
- Static methods cannot be overridden (only hidden)
- Static import allows direct access of static members